

Tier- Nested Shell Composition for Nominal Operator Architecture

A unified syntactic shell model integrating MGD, HTQA tier nesting, and RED/DED traversal

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Abstract

This chapter specifies a syntactic (functional) shell- composition architecture for the nominal phrase and aligns it with three programmatic components developed in the companion manuscripts: (i) Modifiability- Graph Diagnostics (MGD) for tier assignment, (ii) a ranked operator/quantifier architecture (HTQA tier nesting) for closure and traversal, and (iii) a bidirectional RED/DED traversal methodology for diagnosing packaging dependencies. The architectural commitment is deliberately minimal and strictly tier- based: Shell 1 hosts LOW primitives; Shell 2 hosts MIDDLE packaging/repair orbitals; Shell 3 hosts HIGH operator ranks; Shell 4 hosts connective bridges; Shell 5 hosts outer polarity and outer minimizers that scope over the assembled closure stack. The model is evaluated not by category labels alone but by graph position and closure sensitivity: shells are empirically testable via directed head- modification graphs and by explicit falsification templates. The chapter integrates cross- linguistic packaging strategies (article, article- less, classifier systems) and provides a filled- slot inventory that is intentionally extensible.

Keywords: nominal architecture; determiners; operators; quantifiers; modifiability graphs; repair; HTQA; RED/DED; closure; polarity.

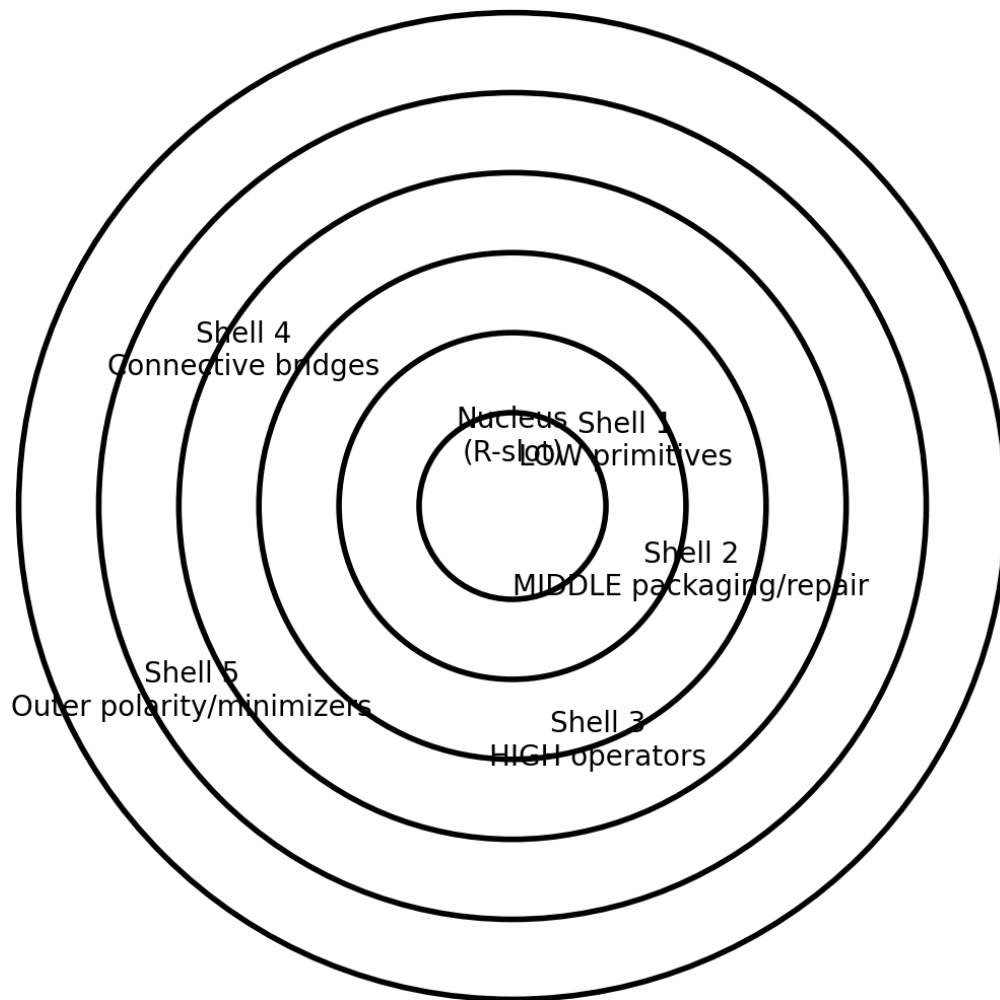
13.1 Scope and Design Commitments

This chapter is intentionally restricted to syntactic shell composition and the associated diagnostic machinery. No independent lexical- ontological genesis model is assumed here; the chapter begins at the level at which nominal structure is already a compositional object and evaluates the internal architecture of that object. The background frameworks are familiar: DP- style nominal functional structure and compositional semantics provide a baseline (Abney 1987; Heim & Kratzer 1998). The contribution of the present system is not the claim that nominal phrases have functional structure, but a testable tier partition with a graph- based diagnostic and an explicit packaging interface that feeds ranked operator closure.

The sole architectural commitments are the five shell roles below. Shell membership is defined by tier locus and closure sensitivity, operationalized by Modifiability- Graph Diagnostics (MGD) and by RED/DED traversal.

Five shell roles (tier- nested): Shell 1 hosts LOW primitives; Shell 2 hosts MIDDLE packaging/repair orbitals; Shell 3 hosts HIGH operator ranks; Shell 4 hosts connective bridges; Shell 5 hosts outer polarity and outer minimizers that scope over the assembled closure stack.

Figure 13.1. Concentric shell view (Nucleus + Shells 1–5).



13.2 The Five-Shell System

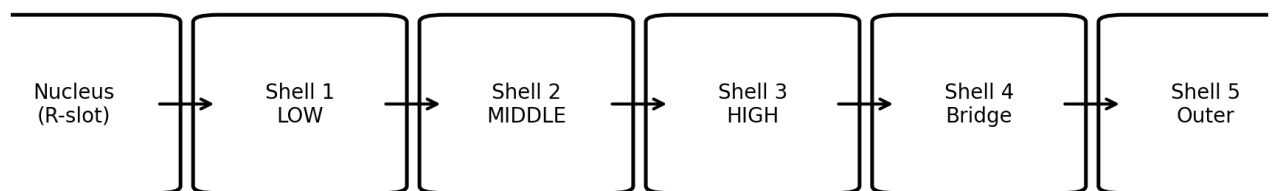
The electron- shell metaphor used throughout the project is operationalized as a tier- nested composition principle. Each shell is a bounded locus with (i) a characteristic inventory, (ii) a characteristic mode of interaction with inner material, and (iii) a characteristic diagnostic signature in MGD. The shells are not defined “by meaning classes”; rather, they are defined by graph position (shell/subshell/orbital) and by closure sensitivity, i.e., by what the head can modify and what it requires to be licensed (Project manuscripts, 2026).

A central empirical claim is the tier profile: LOW is sparse (isolate- heavy), MIDDLE is structurally connective (SCC witnesses), HIGH is isolate- dominant under strict head- modification, Bridge material links blocks without participating in tier- internal head modification, and Outer polarity/minimizers are scope- dominant (Project manuscripts, 2026).

Table 13.1. Shell inventory, tier affiliation, and diagnostic expectations.

Shell	Tier	Core function	MGD signature (expected)	Representative inventory (English; working)
Nucleus	—	Nominal anchor (R- slot)	Target of restriction; input to closure	N; deverbal nominals; proper- name stems
Shell 1	LOW	Primitives and noun- slot substitutes	Sparse edges; isolate- heavy	each other; who/what; free relatives (what(you said), whoever)
Shell 2	MIDDLE	Packaging & repair orbitals	Nontrivial SCC witnesses; moderate density	ellipsis; pronounization; one- support; classifier/measure support; coercion
Shell 3	HIGH	Ranked operators; closure & traversal	Isolate- dominance under strict criterion	almost; all/both; the/a; every/each/no; absolutely; frankly
Shell 4	Bridge	Connective/comparative/partitive linking	Links blocks; no tier- internal head modification	than/as; and/or/but; of; with/for/in/without
Shell 5	Outer	Polarity & minimizers with wide scope	Scope- dominant; outer constraints	not/n't/never; hardly/scarcely/barely

Figure 13.2. Wrap procedure: stabilization from Nucleus to outer shells.



Wrap procedure: each shell stabilizes the inner object before the next shell applies.

The wrap procedure is a methodological constraint: each shell must be able to form a stable intermediate object before the next shell applies. The constraint eliminates “short- circuiting” analyses where outer heads appear to modify inner material directly but are in fact scoping over a phrase. As a result, Shell 2 becomes the unique locus for supplier recruitment and repair (ellipsis, pronounization, one- support, classifier/measure insertion), and Shell 3 becomes the unique locus for ranked closure and traversal control (Project manuscripts, 2026).

13.2.1 Worked English Examples: Shell-Distributed Derivations

This subsection is added as a worked demonstration layer for the five- shell methodology. It does not revise any definitions above; it instantiates them with fully spelled- out English derivations that make shell membership observable in the evidence object used by Modifiability- Graph Diagnostics. The examples are organized as step- wise wraps from the nucleus outward (cf. Abney 1987; Heim & Kratzer 1998; and the project’s MGD and Electron- Shell chapters).

Example set A: qualitative stacking (“the very dark red rose”)

Target string: the very dark red rose. Interpretive goal: demonstrate that qualitative property heads package in Shell 2, that degree/intensifier heads occupy Shell 3 (HIGH operator ranks), and that D- closure likewise occupies Shell 3, while Shell 4/5 remain empty in this minimal case.

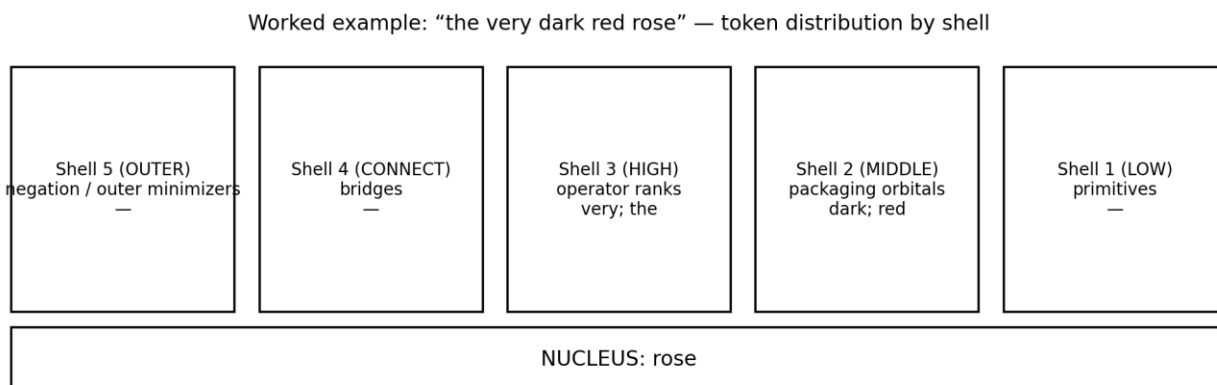


Figure 13.2a. Shell distribution for “the very dark red rose” (worked illustration).

Derivation (wrap steps)

- Step 0 (Nucleus): rose
- Step 1 (Shell 2, qualitative): red → rose
- Step 2 (Shell 2, qualitative refinement): dark → red (preserving red→rose)
- Step 3 (Shell 3, HIGH operator): very → dark (preserving dark→red→rose)
- Step 4 (Shell 3, closure): the → rose (closure targets the nominal head; internal modifiers remain below)

MGD note: Under strict head- modification, the recorded edges are the head- targeting links very→dark, dark→red, red→rose, and the→rose. No Shell- 4 connector is required in this example; no Shell- 5 outer polarity is present. The distribution is therefore a direct diagnostic of tier membership by graph position (project drafts: Modifiability- Graph Diagnostics; Chapter 12 Electron- Shell).

Table 13.2a. Token - to - shell mapping for “the very dark red rose”.

Token	Shell	Orbital (if applicable)	Head- target	Edge recorded by MGD
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rose	Nucleus	—	—	—
red	Shell 2 (MIDDLE)	Qualitative	rose	red→rose
dark	Shell 2 (MIDDLE)	Qualitative	red	dark→red
very	Shell 3 (HIGH)	Operator (degree/intensifier)	dark	very→dark
the	Shell 3 (HIGH)	Closure (D- rank)	rose	the→rose

Example set B: quantitative + operator stacking (“almost all three roses”)

Target string: almost all three roses. Purpose: demonstrate the separation of a quantitative packaging head (numeral) in Shell 2 from operator ranks in Shell 3 (proximity ‘almost’ + pre- closure universal ‘all’). This yields the characteristic HIGH- tier profile (operator isolate tendency with constrained operator- operator modification: almost→all) and stable operator- to- nominal links (all→roses).

Derivation (wrap steps)

- Step 0 (Nucleus): roses
- Step 1 (Shell 2, quantitative): three → roses
- Step 2 (Shell 3, pre- closure universal): all → roses
- Step 3 (Shell 3, proximity): almost → all

13.2.2 Cross-Linguistic Micro-Dataset: Same Shells, Different Linearization

This subsection supplies a minimal cross- linguistic dataset to match the project’s cross- linguistic orientation. The methodological point is invariant: shell membership is determined by tier locus and closure role, not by surface order. Languages vary in adjective placement, the presence/absence of overt articles, linker morphology, and classifier/measure resources. These differences are treated as linearization and connector choices over the same five- shell functional inventory (Abney 1987; Heim & Kratzer 1998; project drafts).

Table 13.2b. Cross - linguistic examples mapped to shells (illustrative; replace with attested citations as desired).

Language	Example	Gloss (informal)	Shell 2 (MIDDLE)	Shell 3 (HIGH)	Shell 4/5 notes
English	the very dark red rose	D + degree + Adj + Adj + N	dark; red	the; very	—
German	die sehr dunkelrote Rose	D + very + dark- red +	dunkelrot(e)	die; sehr	—

		rose			
Spanish	la rosa muy roja	D + rose + very + red	roja	la; muy	Adj post- N
Russian	очень темно- красная роза	very + dark- red + rose	темно- красная	очень (D is Ø)	no overt article
Turkish	çok koyu kırmızı gül	very + dark + red + rose	koyu; kırmızı	çok (D is Ø)	Adj pre- N
Mandarin	nà duǒ fēicháng shēn hóng de méiguī	that CL very deep red DE rose	shēn hóng	fēicháng; nà	de linker (CONNECT); CL measure/interface

Cross- linguistic diagnostic note: (i) article- less languages may realize Shell- 3 closure as Ø while still showing Shell- 3 operator behavior via degree/proximity items; (ii) classifier languages externalize grouping resources in Shell- 2 measure/interface orbitals; (iii) linker morphology (e.g., Mandarin de) behaves as Shell- 4 CONNECT material bridging modifiers to the nominal core.

13.2.3 Formal Core + Results (Definitions and Architectural Consequences)

This block is inserted here—immediately before the MGD procedure section—so that the remainder of the chapter can treat Modifiability- Graph Diagnostics as a formally defined evidence object and can state the shell architecture as a system with consequences. The content below uses only the shell/tier commitments already introduced in this chapter (Shells 1–5; wrap procedure; tier signatures; interface supply/retargeting; RED/DED).

13.2.3.1 Formal Core: Objects, Notation, and Evidence Type

This subsection states the shell architecture as a formal object that can be inferred and tested using the project’s Modifiability- Graph Diagnostics (MGD). The aim is not to reintroduce background DP/QP assumptions (see Abney 1987; Heim & Kratzer 1998 for standard baselines), but to define the evidence type that the present chapter treats as primary: tier- indexed directed head- modification graphs plus wrapper traversal procedures (RED/DED).

Definition 1 (Shell-indexed inventory).

Let V be the set of nominal- internal heads considered in the analysis (lexical and functional heads that can act as heads under the strict head- targeting criterion). A shell assignment is a total function $\sigma: V \rightarrow \{1,2,3,4,5\}$, where: $\sigma(x)=1$ denotes Shell 1 (LOW) primitives; $\sigma(x)=2$ denotes Shell 2 (MIDDLE) packaging orbitals; $\sigma(x)=3$ denotes Shell 3 (HIGH) operator ranks (closure/entry control); $\sigma(x)=4$ denotes Shell 4 (CONNECT) bridges; and $\sigma(x)=5$ denotes Shell 5 (OUTER) outer polarity and outer minimizers that scope over the assembled closure stack. Write $V_{\square} = \{x \in V \mid \sigma(x)=k\}$ for the induced partition.

Definition 2 (Strict head -modification relation and MGD graph).

MGD represents nominal composition by a directed relation $\rightarrow \subseteq V \times V$ interpreted as strict head- targeting modification. An edge $x \rightarrow y$ is recorded iff head x modifies head y (not merely an XP containing y), under the strict head- targeting criterion. The MGD graph is $G=(V,E)$ where $E=\{(x,y) \mid x \rightarrow y\}$. For each shell k , the tier subgraph is $G_{\square}=(V_{\square},E_{\square})$ with $E_{\square}=E \cap (V_{\square} \times V_{\square})$. Cross- tier edges (e.g., Shell- 3 closure targeting the nominal head) are retained in G and are used to diagnose wrap order and closure targeting.

Definition 3 (Tier signatures: isolates, SCC witnesses, density).

Given a directed graph $H=(W,F)$: an isolate is a node with zero in- degree and zero out- degree in H ; an SCC witness is a strongly connected component of size ≥ 2 ; and density is $\rho(H)=|F|/(|W|(|W|-1))$ for $|W| \geq 2$. The functional shell commitments stated in this chapter entail characteristic tier signatures: Shell 3 (HIGH) tends toward isolate dominance (operators rarely admit true head- head modification under strict criteria); Shell 2 (MIDDLE) admits SCC witnesses (iterative packaging and retargeting can create cycles); Shell 1 (LOW) is sparse; Shells 4–5 behave as bridging/scope tiers rather than internal head- stacking domains.

Definition 4 (Wrap procedure as tier -ordered construction).

A wrap procedure is an ordered construction sequence that forms a nominal object by successive shell application: Nucleus $\Rightarrow W_1 \Rightarrow W_2 \Rightarrow W_3 \Rightarrow B_4 \Rightarrow O_5$, where W_1 hosts optional Shell- 1 primitives, W_2 wraps Shell- 2 packaging orbitals, W_3 applies Shell- 3 operator ranks for closure/entry control, B_4 introduces Shell- 4 bridges (comparatives/coordination/prepositions/partitives), and O_5 applies Shell- 5 outer scope (polarity and outer minimizers). This is the functional form of the electron- shell layering: tiers are construction stages with predicted graph consequences.

Definition 5 (Interface supply and retargeting).

When strict head- modification would otherwise be undefined (missing head/type), an interface- supply operation introduces a supplied head that becomes the target of subsequent head- modification. Let R be a partial function that, given a configuration C , returns either a supplied head $h \in V_2$ or $\perp$ (“no supply”). Retargeting means that an intended modifier m attaches to the supplied head h rather than to a non- head- complete constituent. This corresponds to the chapter’s interface orbital logic and the diagnostic rule that edges track headable targets rather than surface strings.

13.2.3.2 Result 1: Shell Identifiability from MGD Invariants (Tier Determination)

Claim.

Assume a dataset D of nominal constructions yields MGD graphs under strict head- targeting. If the tier- signature conditions hold robustly across D —(i) HIGH isolate dominance for operator- rank candidates, (ii) MIDDLE SCC witnesses concentrated in packaging/interface zones, (iii) LOW sparsity for primitives, (iv) CONNECT behavior primarily as bridging rather than dense tier- internal stacking, and (v) OUTER behavior as scope over the assembled closure stack—then the shell partition σ is identifiable from the graphs up to isomorphism: there is a uniquely determined tier assignment that best satisfies the signature constraints simultaneously.

Contrast with the standard view.

In standard nominal architectures, tier placement is typically justified by syntactic labels (DP/QP/NegP) and interpretive assumptions; multiple labelings can be made consistent with truth- conditional semantics (Abney 1987; Heim & Kratzer 1998). Here, tier placement is constrained by a recoverable invariant profile of a computed evidence object: isolate rates, SCC distribution, and density patterns in MGD graphs. This yields a representation identifiability claim rather than a descriptive layer list.

Unique prediction.

If Shell 3 corresponds to closure/entry control as stated in this chapter, then across languages/registers items hypothesized as HIGH operators will exhibit systematic isolate tendency under strict head- targeting (apparent “operator modification” resolves as scope- level or Shell- 2 retargeting, not true operator- operator stacking). SCC witnesses should cluster in Shell 2 due to iterative packaging and interface retargeting. Shell 4 should appear chiefly as cross- block bridging rather than as a dense tier- internal head- stacking tier. Operationally, tier summary statistics separate tiers: $\text{Iso}(G_3)$ high; $\text{SCC}(G_2)$ nontrivial; $\rho(G_1)$ low.

What would falsify it.

A robust falsifier is a dataset where, under strict head- targeting (not scopal paraphrase), the candidate HIGH operator set forms reliable SCCs (true operator- operator mutual modification) with repeated witnesses; packaging/interface zones do not concentrate SCC behavior; or Shell- 4 connectors behave as dense tier- internal modifiers rather than bridges. Any of these patterns would break the claimed identifiability because the invariants would no longer carve the tiers cleanly.

13.2.3.3 Result 2: No Short-Circuit Closure (Wrap Necessity for Closure Targeting)

Claim.

In the five- shell wrap system, a Shell- 3 closure head cannot target a configuration that has not achieved head- complete packaging in Shell 2 without either (i) triggering interface supply/retargeting in Shell 2, or (ii) yielding failure of strict head- targeting (no licit edge set under MGD). Informally: closure cannot short- circuit packaging.

Contrast with the standard view.

In label- based accounts, it is often possible to merge a closure head with an NP and let semantics resolve well- formedness without a structural necessity claim. Under the present shell system, closure is a tier whose functional role is to cap internal modification and impose entry/closure constraints; that role presupposes a packaged object with a stable head target. Therefore, closure is predicted to be dependent on prior packaging stages.

Unique prediction.

Where a nominal restrictor is not head- complete, closure heads will be acceptable only if an interface supplier provides a head target (yielding observable retargeting in MGD). Where interface supply is blocked, closure readings degrade: acceptability drops or the reading shifts to a different scope configuration. In graph terms, attempted closure over unpackageable material

yields either missing edges (MGD failure) or replacement edges to the supplied head (retargeting), not stable direct closure edges to a non-head-complete target.

What would falsify it.

A falsifier is a systematic pattern where closure heads repeatedly and robustly form strict head-targeting edges to structures classified as not head-complete under Shell 2, with no evidence of interface supply/retargeting (no supplied head, no repair footprint), and where the pattern is stable across speakers/corpora. Such cases would show that closure can short-circuit packaging, contradicting the result.

13.2.3.4 Result 3: Non-Reducibility to Label-Only Architectures (Graph-Visible Distinctions)

Claim.

There exist nominal configurations for which multiple label-only representations (DP/QP-style) are truth-conditionally equivalent but yield different head-targeting dependencies. The five-shell system predicts that these representations are not equivalent because MGD distinguishes them by graph topology. Therefore, the shell theory is not a mere relabeling of DP/QP; it adds a graph-visible structural dimension.

Contrast with the standard view.

In standard compositional semantics, many differences between head attachment and phrasal adjunction can be made truth-conditionally inert (Heim & Kratzer 1998). Label-based syntax can therefore tolerate multiple LFs that compute the same truth conditions. The present shell system commits to strict head-targeting as the diagnostic evidence type and treats tier membership as a property of where head-head edges actually occur.

Unique prediction.

When two analyses are truth-conditionally equivalent but differ in whether an element is a head-modifier vs a phrase-level adjunct, the shell system predicts that the analysis yielding strict head-targeting edges consistent with the tier signatures is architecturally correct. The alternative analysis will predict the wrong tier signature: it will erase expected cross-tier closure links, alter isolate counts, or destroy SCC witnesses where packaging requires them.

What would falsify it.

This claim is weakened if one can reproduce the full tier signatures—HIGH isolate dominance, MIDDLE SCC witnesses, LOW sparsity, CONNECT bridging, OUTER scope—without committing to strict head-targeting edges, i.e., if a purely label-based DP/QP architecture with flexible phrasal adjunction can match the observed graph invariants across datasets while remaining truth-conditionally equivalent.

13.3 Modifiability-Graph Diagnostics (MGD) for Tier Assignment

13.3.0 Micro-Evidence Layer: Reference-Attested Tier Graph Counts (MGD)

To minimally instantiate the predicted tier signatures with numbers (rather than schematic diagrams alone), we reproduce the tier-internal graph statistics from the reference-attested

instantiated graphs reported in the Tier- Nesting chapter (§5.1 “Graph statistics”). These counts are based on conservative Tier- 1 coding and the strict head- targeting edge criterion. The goal is not broad corpus coverage; it is a small, auditable numerical witness that the architecture yields the expected contrast: HIGH is isolate- dominant while MIDDLE supports a nontrivial SCC witness (cf. Abney 1987; Heim & Kratzer 1998).

Table 13.3 - A. Reference - attested tier statistics under strict head - modification (MGD).

Tier	#Nodes	#Edges	Density	Isolates	SCCs (size>1)	Salient witness notes
HIGH	10	0	0.000	THE; A/AN; Ø; ALL; EVERY; EACH; NO; ALMOST; ABSOLUTELY; NOT	—	Operator heads behave as isolates under strict head- modification.
MIDDLE	11	4	0.036	ELL_NP; PRON; CLASSIFIER/PROP- NO UN; COERCE; EXCEPTIVE_BUT	{SORT_OF, KIND_OF}	SCC witness persists under strict criterion; OTHER→ONE illustrates packaging.
LOW	7	4	0.095	RECIP_EACH_OTHER; FREE_REL	—	Sparse; dominated by restricted modifier- clause → nominal- head relations.

Derived contrast (immediate signature). Under these instantiated graphs, HIGH has 10/10 isolates (1.00), while MIDDLE is only partially isolate with a nontrivial SCC witness {SORT_OF, KIND_OF}; LOW remains sparse. These are the tier- signature contrasts the present chapter treats as diagnostic objects.

13.3.0.1 Second thin increment: Tier statistics on the Chapter 13 example set

As a second minimal numerical witness using only the chapter’s worked English and cross- linguistic examples (§13.2.1–13.2.2), we build strict head- modification edges for each example and then aggregate tier- internal edges by shell. This does not introduce new inventory commitments; it merely converts the example set into the same statistics reported in Table 13.3- A.

Table 13.3 - B. Tier statistics computed from the Chapter 13 English + cross - linguistic example set.

Tier	#Nodes	#Edges	Density	Isolates	SCCs (size>1)	Notes
HIGH (Shell 3)	12	1	0.008	the; very; die; sehr; la; muy; očen; çok;	—	Example- set HIGH shows isolate dominance

				fěicháng; nà		with only one operator- operator edge (almost→all).
MIDDLE (Shell 2)	10	2	0.022	three; dunkelrot(e); roja; temno- krasnaja; shēn hóng; duǒ	—	MIDDLE shows limited internal edges from adjectival stacking; SCC witnesses are not expected in minimal adjectival sets.
LOW (Shell 1)	0	0	0.000	—	—	No LOW- tier primitives were included in the minimal worked example set; LOW statistics are therefore empty here.

Interpretation. Even on the small worked example set, the tier contrast holds: HIGH is overwhelmingly isolate- dominant (most HIGH nodes participate in no tier- internal edges), while MIDDLE is more connected due to modifier stacking. The SCC witness signature characteristic of MIDDLE is not forced by adjectival stacks alone; it arises in the packaging/interface inventory (as in Table 13.3- A’s {SORT_OF, KIND_OF} witness) and is therefore predicted to appear when those packaging items are included in the sample.

MGD is the principal empirical diagnostic for the tier system. For a fixed language L and shell/tier k, MGD constructs a directed graph $G_{\{L,k\}} = \langle V_{\{L,k\}}, E_{\{L,k\}} \rangle$, where nodes are tier inventory heads and an edge $X \rightarrow Y$ is recorded iff X directly modifies Y as a head under a strict criterion. The strict criterion intentionally excludes relations that are purely scopal, purely metalinguistic, or mediated by covert structural repair. When repair is required, the diagnostic targets the supplied head (ELL/PRON/REPAIR/COERCE) rather than the surface trigger (Project manuscripts, 2026).

Graph topology is the evidence object: isolates quantify “non- stacking” behavior; strongly connected components (SCCs) witness iterative packaging; and density separates LOW sparsity from MIDDLE connectivity and HIGH isolate dominance. These signatures are predictions, not stipulations: they can be supported by corpus counts and by acceptability- judgment grids, and they can be falsified by counterexample patterns (Section 13.10).

Table 13.2. MGD DB- 0 protocol and minimal reporting fields.

Component	Specification
DB- 0 protocol	1) Fix tier inventory V. 2) Propose candidate edges from minimal pairs, elicitation, or corpus. 3) Apply strictness filter (head

	targeting; exclude scope-only). 4) Control ellipsis/pro- drop. 5) If licensing requires repair (ELL/PRON/REPAIR/COERCE), target the supplied head. 6) Compute V , E , density, isolates, SCCs; record witness items.
Evidence (τ)	REF- ATTESTED / CORPUS- ATTESTED / ELICITED
Judgment (j)	A / M / C / E / U
Repair codes	ELL / PRON / REPAIR / COERCE

Figure 13.3. Schematic modifiability graphs by tier (strict head- modification).

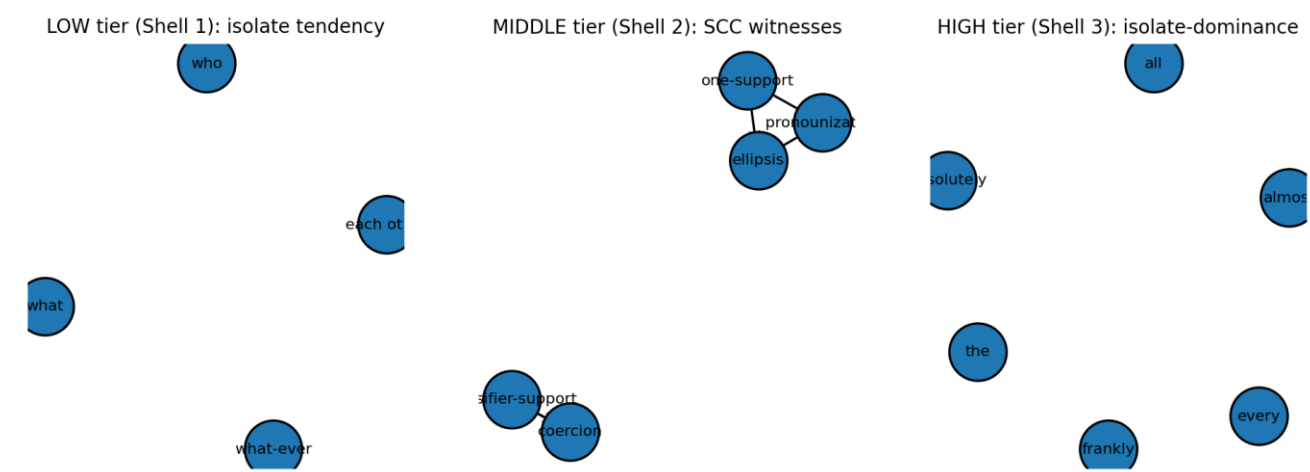
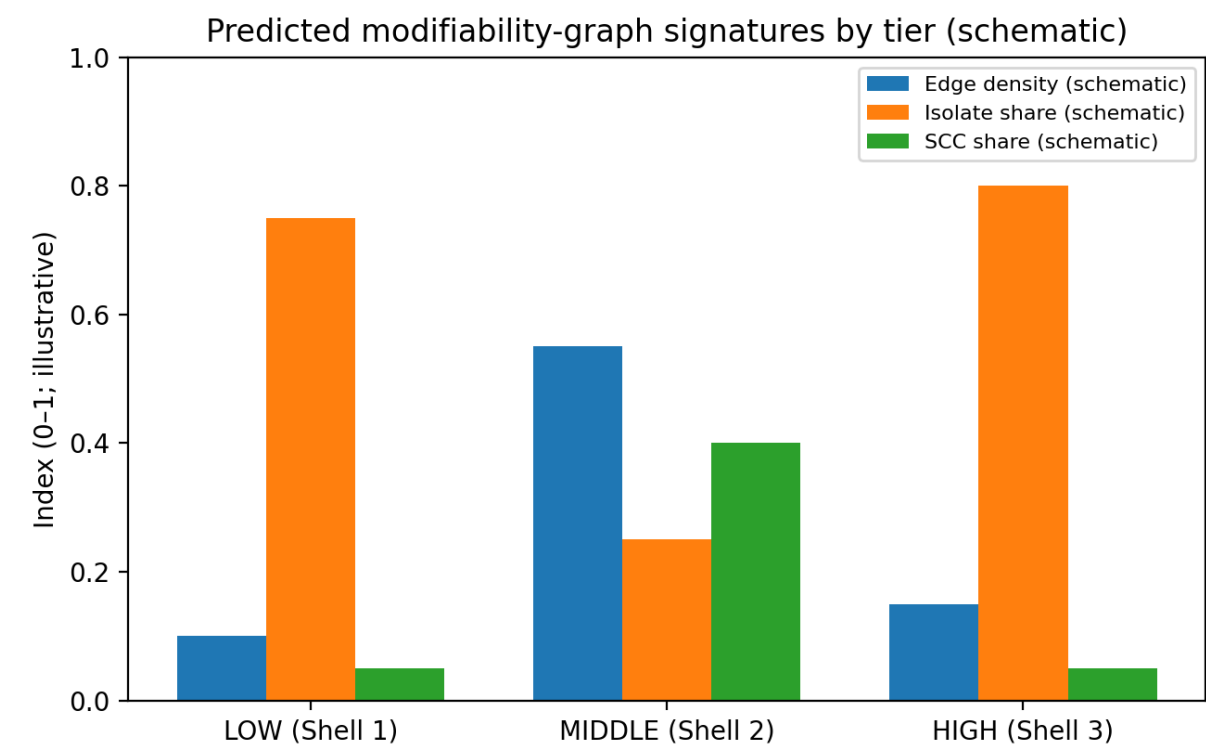


Figure 13.4. Predicted tier signatures (schematic indices; illustrative).



13.4 Shell 1 (LOW): Primitives and Noun-Slot Substitutes

Shell 1 hosts LOW primitives: items that can substitute for a nominal slot or introduce a minimal nominal argumental configuration without triggering packaging repair. The Shell- 1 class is diagnostic in MGD because its internal head- modification potential is limited: it is expected to be isolate- heavy and sparsely connected. This is compatible with the observation that many Shell- 1 items are licensed by clause- level dependencies (e.g., interrogation) or by anaphoric binding, rather than by NP- internal packaging mechanics (Project manuscripts, 2026).

Working Shell- 1 inventory includes: reciprocals (each other), nominal wh- forms (who, what), and free- relative/ever- forms (whoever, whatever). These items serve as direct NP substitutes and often impose their own external licensing conditions (Hamblin 1973; Karttunen 1977; Project manuscripts, 2026).

Table 13.3. Shell 1 (LOW) inventory (working list).

Subclass	Examples	Diagnostic notes (strict head- modification)
Reciprocal primitive	each other	NP substitute; binding- licensed; isolate tendency
Nominal wh- substitute	who, what	Clause- licensed; may not admit tier- internal modification
Ever/free relatives	whoever, whatever; what(you said)	NP substitute; external licensing; isolate tendency

13.5 Shell 2 (MIDDLE): Packaging/Repair Orbitals

Shell 2 is the hybrid transition zone. It is where descriptive restriction is built into quantitative structure and where interface mechanisms deliver a closure- ready package to the HIGH operator layer. Shell- 2 material is predicted to show nontrivial connectivity under MGD, including SCC witnesses, because packaging and repair can apply iteratively and can target outputs of earlier packaging steps (Project manuscripts, 2026).

The electron- shell refinement decomposes Shell 2 into three orbitals ordered by increasing packaging complexity: qualitative restriction (2.1), quantitative structuring (2.2), and interface repair/supplier recruitment (2.3). These orbitals correspond to distinct graph subregions and can be separately evaluated.

Table 13.4. Shell 2 orbitals (qualitative → quantitative → interface) with typical fillers (working list).

Orbital	Role	Typical fillers
2.1 Qualitative	Restriction building near nucleus	N- modifier; adjectives; comparative/superlative morphology; degree: very, extremely, quite, rather, too,

		enough, so, as
2.2 Quantitative	Counting/ordering and magnitude assessment	cardinals; ordinals; measure/classifier phrases; some/many/few/a few/several; approximators: about/around/nearly
2.3 Interface	Supplier recruitment + repair	ellipsis; pronounization; prop- noun one; classifier/measure support; coercion/nominalization; anaphoric packaging

13.6 Shell 3 (HIGH): Ranked Operator Architecture

Shell 3 hosts ranked operator heads that control closure and traversal. The operator/quantifier distinction is maintained as a structural distinction: many quantificational expressions can be treated as generalized quantifiers in the semantics (Barwise & Cooper 1981; Keenan & Stavi 1986), but only a subset behave as NP heads without repair. Operators are defined by closure sensitivity: they require a package that has been stabilized by Shell- 2 orbitals and they contribute scopal/traversal control rather than NP- head saturation (Project manuscripts, 2026).

Under MGD's strict head- modification criterion, Shell- 3 is predicted to be isolate- dominant: operators rarely form genuine head- modification chains. Apparent stacking is instead captured by rank ordering, selectional compatibility, and closure features. This yields a controllable inventory and precise falsification targets: a stable HIGH SCC with strict head- targeting witnesses would directly challenge the rank- based account (Section 13.10).

Table 13.5. Shell 3 operator ranks (filled inventory; working list).

Rank	Domain	Examples	Structural role
O4- a	Proximity	almost, nearly	Thresholding before closure
O4- b	Pre- closure universal	all, both	Universal scope before closure
O4- c	Closure (D/Q)	the, a/an, Ø; this, that; every, each, no; to (non- finite marker)	Gatekeeping closure; traversal readiness
O4- d	Maximizers	absolutely, totally, completely, entirely, utterly, perfectly	Endpoint saturation
O4- e	Modal/Evaluative	frankly, allegedly, theoretically, surprisingly	Speaker stance/evaluation

Quantifier heads (as opposed to higher operators) are typically realized in Shell 2 quantitative orbitals: they head the nominal package and can saturate nominal requirements without supplier recruitment. Operators (especially closure ranks) are typically realized in Shell 3. This division supports typological variation: languages may lack articles but maintain packaging and closure

via classifiers, possessive morphology, or alternative determiners (Haspelmath 1997; Project manuscripts, 2026).

Table 13.6. Structural diagnostics: quantifier heads vs operator heads.

Property	Quantifier heads (Shell 2.2)	Operator heads (Shell 3)	Illustrative English patterns
Headability	Can function as NP head	Cannot head NP without supplier/repair	Many left vs *No left → None left / No one left
Repair dependence	Often independent; may recruit measure packaging	Frequently depends on closure- ready package	the red one; those; none
MGD signature	MIDDLE connectivity possible	HIGH isolate dominance	SCC witnesses vs isolates
Scope behavior	Typically NP- internal restriction/amount	Controls closure/traversal; can take wide scope	every N; no N; the N

13.7 Cross-Linguistic Packaging Strategies within the Same Shell Skeleton

The shell system is intended to be cross- linguistically extensible. Variation is expected in (i) which inventories are overt, (ii) which Shell- 2 suppliers are preferred (prop- noun vs classifier vs anaphor), and (iii) how Shell- 3 closure is morphologically expressed (article, clitic, affix, or zero). The shell skeleton itself remains constant: LOW substitutes, MIDDLE packaging, HIGH closure/operators, bridging connectors, and outer polarity/minimizers (Project manuscripts, 2026).

Table 13.7. Illustrative packaging strategies (schematic, aligning to shells).

System type	Typical overt strategy	Shell locus	Illustrative pattern
Article language	Articles/determiners as closure heads	Shell 3 (O4- c)	English the/a; demonstratives this/that
Classifier language	Classifier/measure as supplier/packaging	Shell 2.3 (Interface)	Mandarin nǎ ge ‘which CL’ (Project manuscripts, 2026)
Article- less language	Closure via morphology/word order/zero	Shell 3 (zero closure)	Russian bare NP with operator effects via context (Project manuscripts, 2026)
Clitic/affixal definiteness	Definite marker attached to noun	Shell 3 (closure morphology)	Arabic al- ; Persian ezafe- mediated packaging (Project manuscripts, 2026)

13.8 Shell 4 (Bridge) and Shell 5 (Outer) in the Diagnostics

Shell 4 hosts bridge material—comparatives, coordination, relational prepositions, and partitive linking. Under the strict head- modification criterion, Shell- 4 items do not participate in tier- internal head modification within Shells 1–3; they link packages. Shell 5 hosts outer polarity and minimizers with wide scope. Shell- 5 items are diagnosed by scope dominance and by restrictions on tier- internal head targeting. These loci are crucial for preserving the internal diagnostic path: MGD graphs are built primarily within Shells 1–3, while Shells 4–5 are modeled as higher- order links/operators over assembled packages (Project manuscripts, 2026).

Table 13.8. Shell 4 bridge inventory (filled working list).

Type	Examples	Role
Comparatives	than; as ... as	Bridge two packages for comparison
Coordination	and; or; but; yet	Conjoin packages
Relational prepositions	with; for; in; without	Relational linking
Partitive bridge	of	Partitive measure/quantity linking
Correlative coordination	both...and; either...or; neither...nor	Bridge with operator- like constraints

Table 13.9. Shell 5 polarity/minimizers (filled working list).

Class	Examples	Notes
Negation	not; n't; never	Polarity reversal; wide scope
Outer minimizers	hardly; scarcely; barely	Thresholding; treated as outer unless clearly scalar- internal

13.9 RED/DED as Bidirectional Shell-Traversal Diagnostics

RED/DED is incorporated as a bidirectional traversal procedure over the shell stack. DED (forward) models constructive assembly from inner material to closure, bridge, and polarity. RED (reverse) is a diagnostic back- tracking procedure: given a surface package, identify the minimal shell locus that must have supplied required heads (supplier recruitment, repair, closure). The core benefit is methodological: RED/DED localizes dependencies (e.g., why an operator is non- headable without an interface supplier) and prevents accidental analyses that attribute packaging to the HIGH tier.

Table 13.10. RED/DED templates (schematic).

Observed pattern	DED assembly	RED diagnostic
*No left	Recruit supplier in Shell- 2 interface → close with Shell- 3 operator	Operator non- headability implies Shell- 2 supply (None/No one)
Mandarin nǎ ge	Shell- 2 classifier packaging → interrogative selection	Classifier indicates Shell- 2 interface orbital
absolutely every N	NP restriction → closure → maximizer	Maximizer implies prior operator rank; diagnose stacking constraints

the red $\emptyset$	Shell- 2 ellipsis repair stabilizes NP	Ellipsis indicates Shell- 2 supplier; closure can then apply
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13.10 Predictions and Falsification Templates

The shell- tier system makes falsifiable commitments. The strongest predictions are: (i) HIGH isolate dominance under strict head- modification; (ii) at least one MIDDLE SCC witness due to iterative packaging/repair; and (iii) repair locality—supplier recruitment is concentrated in Shell 2. Each prediction has a minimal falsifier schema.

Table 13.11. Minimal falsifiers for the shell- tier model.

Claim	Unique prediction	Minimal falsifier	Revision target
HIGH isolate signature	Shell- 3 operators remain isolate- dominated	Stable HIGH SCC with strict head- targeting witnesses	Rank system / strictness rubric; fusion vs stacking
MIDDLE SCC witness	Shell- 2 graphs contain SCC witnesses	No SCCs across inventories despite iterative packaging	Orbital decomposition; inventory granularity
Repair locality	Operators cannot create NP heads outside suppliers	Operator becomes headable when ellipsis/pronounization excluded	Supplier diagnostics; repair typing

13.11 Conclusion

The chapter delivered a strictly functional shell- composition architecture for nominal phrases: five shells mapped to a tier- nested system and evaluated with a reproducible graph- theoretic diagnostic. Shell 1 contributes LOW primitives; Shell 2 mediates packaging and repair through orbitals; Shell 3 implements ranked operator closure and traversal; Shell 4 supplies connective bridges; Shell 5 supplies outer polarity/minimizer scope. The system is designed to be extensible: inventories can be enriched cross- linguistically while the diagnostic object (tier- specific modifiability graphs and falsification templates) remains stable.

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